

УО «Белорусский государственный университет информатики и
радиоэлектроники»
Кафедра ПОИТ

Отчет по лабораторной работе №1
по предмету
Распределенные вычисления
Вариант 1

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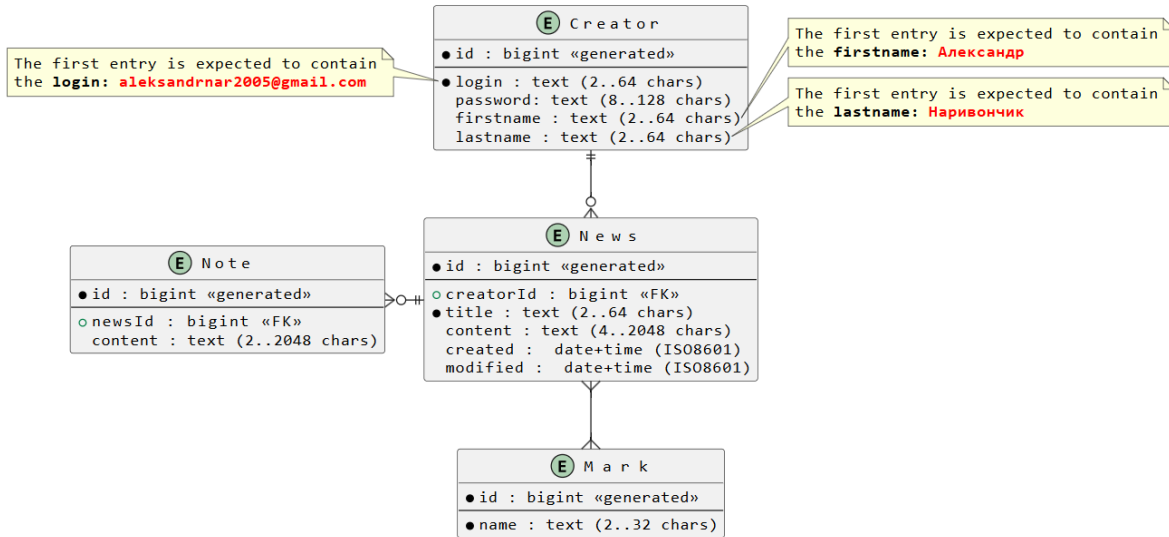
Группа 351004

Минск 2026

Задание

Задание

- Основой для задания является следующая схема приложения:



Код

```
using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

namespace rest1.api.controllers;

[ApiController]
[Route("api/v1.0/[controller]")]
public class CreatorsController : ControllerBase
{
    private readonly ICreatorService _creatorService;
    private readonly ILogger<CreatorsController> _logger;

    public CreatorsController(ICreatorService creatorService,
        ILogger<CreatorsController> logger)
    {
        _creatorService = creatorService;
        _logger = logger;
    }
}
```

```

    }

    [HttpPost]
    [ProducesResponseType(typeof(CreatorResponseTo),
    StatusCodes.Status201Created)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<CreatorResponseTo>> CreateCreator(
        [FromBody] CreatorRequestTo createCreatorRequest)
    {
        try
        {
            _logger.LogInformation("Creating creator {request}",
            createCreatorRequest);

            CreatorResponseTo createdCreator =
                await _creatorService.CreateCreator(createCreatorRequest);

            return CreatedAtAction(
                nameof(CreateCreator),
                new { id = createdCreator.Id },
                createdCreator
            );
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error creating creator");
            return StatusCode(500, "An error occurred while creating the creator");
        }
    }

    [HttpGet]
    [ProducesResponseType(typeof(IEnumerable<CreatorResponseTo>),
    StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<IEnumerable<CreatorResponseTo>>>
    GetAllCreators()
    {
        try
        {

```

```

        _logger.LogInformation("Getting all creators");

        IEnumerable<CreatorResponseTo> creators =
            await _creatorService.GetAllCreators();

        return Ok(creators);
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error getting all creators");
        return StatusCode(500, "An error occurred while retrieving creators");
    }
}

[HttpGet("{id:long}")]
[ProducesResponseType(typeof(CreatorResponseTo),
    StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<CreatorResponseTo>>
    GetCreatorById([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Getting creator by id: {Id}", id);

            var getCreatorRequest = new CreatorRequestTo { Id = id };
            CreatorResponseTo creator =
                await _creatorService.GetCreator(getCreatorRequest);

            return Ok(creator);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "Creator not found with id: {Id}", id);
            return NotFound($"Creator with id {id} not found");
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting creator by id: {Id}", id);

```

```

        return StatusCode(500, "An error occurred while retrieving the creator");
    }
}

[HttpPut]
[ProducesResponseType(typeof(CreatorResponseTo),
StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status400BadRequest)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<CreatorResponseTo>> UpdateCreator(
    [FromBody] CreatorRequestTo updateCreatorRequest)
{
    try
    {
        _logger.LogInformation(
            "Updating creator with id: {Id}",
            updateCreatorRequest.Id);

        var updatedCreator =
            await _creatorService.UpdateCreator(updateCreatorRequest);

        return Ok(updatedCreator);
    }
    catch (NewNotFoundException ex)
    {
        _logger.LogWarning(
            ex,
            "Creator not found for update with id: {Id}",
            updateCreatorRequest.Id);

        return NotFound(
            $"Creator with id {updateCreatorRequest.Id} not found");
    }
    catch (Exception ex)
    {
        _logger.LogError(
            ex,
            "Error updating creator with id: {Id}",
            updateCreatorRequest.Id);
    }
}

```

```

        return StatusCode(
            500,
            "An error occurred while updating the creator");
    }
}

[HttpDelete("{id:long}")]
[ProducesResponseType(StatusCodes.Status204NoContent)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<IActionResult> DeleteCreator([FromRoute] long id)
{
    try
    {
        _logger.LogInformation("Deleting creator with id: {Id}", id);

        var deleteCreatorRequest = new CreatorRequestTo { Id = id };
        await _creatorService.DeleteCreator(deleteCreatorRequest);

        return NoContent();
    }
    catch (CreatorNotFoundException ex)
    {
        _logger.LogWarning(
            ex,
            "Creator not found for deletion with id: {Id}",
            id);

        return NotFound();
    }
    catch (Exception ex)
    {
        _logger.LogError(
            ex,
            "Error deleting creator with id: {Id}",
            id);

        return StatusCode(
            500,

```

```

        "An error occurred while deleting the creator");
    }
}
}

```

```

using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

```

```

namespace rest1.api.controllers;

```

```

[ApiController]
[Route("api/v1.0/[controller]")]
public class MarksController : ControllerBase
{
    private readonly IMarkService _markService;
    private readonly ILogger<MarksController> _logger;

    public MarksController(IMarkService markService, ILogger<MarksController>
logger)
    {
        _markService = markService;
        _logger = logger;
    }

    [HttpPost]
    [ProducesResponseType(typeof(MarkResponseTo),
StatusCodes.Status201Created)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<MarkResponseTo>> CreateMark(
        [FromBody] MarkRequestTo createMarkRequest)
    {
        try
        {
            _logger.LogInformation("Creating mark {request}", createMarkRequest);

            MarkResponseTo createdMark =

```

```

        await _markService.CreateMark(createMarkRequest);

        return CreatedAtAction(
            nameof(CreateMark),
            new { id = createdMark.Id },
            createdMark
        );
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error creating mark");
        return StatusCode(500, "An error occurred while creating the mark");
    }
}

[HttpGet]
[ProducesResponseType(typeof(IEnumerable<MarkResponseTo>),
    StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<IEnumerable<MarkResponseTo>>>
    GetAllMarks()
    {
        try
        {
            _logger.LogInformation("Getting all marks");

            IEnumerable<MarkResponseTo> marks =
                await _markService.GetAllMarks();

            return Ok(marks);
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting all marks");
            return StatusCode(500, "An error occurred while retrieving marks");
        }
    }

[HttpGet("{id:long}")]
[ProducesResponseType(typeof(MarkResponseTo), StatusCodes.Status200OK)]

```

```

[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<MarkResponseTo>>
GetMarkById([FromRoute] long id)
{
    try
    {
        _logger.LogInformation("Getting mark by id: {Id}", id);

        var getMarkRequest = new MarkRequestTo { Id = id };
        MarkResponseTo mark =
            await _markService.GetMark(getMarkRequest);

        return Ok(mark);
    }
    catch (NewNotFoundException ex)
    {
        _logger.LogWarning(ex, "Mark not found with id: {Id}", id);
        return NotFound($"Mark with id {id} not found");
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error getting mark by id: {Id}", id);
        return StatusCode(500, "An error occurred while retrieving the mark");
    }
}

```

```

[HttpPut]
[ProducesResponseType(typeof(MarkResponseTo), StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status400BadRequest)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<MarkResponseTo>> UpdateMark(
    [FromBody] MarkRequestTo updateMarkRequest)
{
    try
    {
        _logger.LogInformation("Updating mark with id: {Id}",
updateMarkRequest.Id);
    }
}

```

```

        var updatedMark =
            await _markService.UpdateMark(updateMarkRequest);

        return Ok(updatedMark);
    }
    catch (NewNotFoundException ex)
    {
        _logger.LogWarning(
            ex,
            "Mark not found for update with id: {Id}",
            updateMarkRequest.Id);

        return NotFound(
            $"Mark with id {updateMarkRequest.Id} not found");
    }
    catch (Exception ex)
    {
        _logger.LogError(
            ex,
            "Error updating mark with id: {Id}",
            updateMarkRequest.Id);

        return StatusCode(
            500,
            "An error occurred while updating the mark");
    }
}

[HttpDelete("{id:long}")]
[ProducesResponseType(StatusCodes.Status204NoContent)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<IActionResult> DeleteMark([FromRoute] long id)
{
    try
    {
        _logger.LogInformation("Deleting mark with id: {Id}", id);

        var deleteMarkRequest = new MarkRequestTo { Id = id };
        await _markService.DeleteMark(deleteMarkRequest);
    }
}

```

```

        return NoContent();
    }
    catch (MarkNotFoundException ex)
    {
        _logger.LogWarning(ex, "Mark not found for deletion with id: {Id}", id);
        return NotFound();
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error deleting mark with id: {Id}", id);
        return StatusCode(500, "An error occurred while deleting the mark");
    }
}
}

```

```

using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

```

```

namespace rest1.api.controllers;

```

```

[ApiController]
[Route("api/v1.0/[controller]")]
public class NewsController : ControllerBase
{
    private readonly INewsService _newsService;
    private readonly ILogger<NewsController> _logger;

    public NewsController(INewsService newsService, ILogger<NewsController>
logger)
    {
        _newsService = newsService;
        _logger = logger;
    }

    [HttpPost]
    [ProducesResponseType(typeof(NewsResponseTo),

```

```

        StatusCode.Status201Created)]
        [ProducesResponseType(StatusCode.Status400BadRequest)]
        [ProducesResponseType(StatusCode.Status500InternalServerError)]
        public async Task<ActionResult<NewsResponseTo>>
CreateNews([FromBody] NewsRequestTo createNewsRequest)
    {
        try
        {
            _logger.LogInformation("Creating news with title: {Title}",
createNewsRequest.Title);

            var createdNews = await _newsService.CreateNews(createNewsRequest);

            return CreatedAtAction(
                nameof(GetNewsById),
                new { id = createdNews.Id },
                createdNews
            );
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error creating news");
            return StatusCode(500, "An error occurred while creating the news");
        }
    }

[HttpGet]
[ProducesResponseType(typeof(IEnumerable<NewsResponseTo>),
StatusCode.Status200OK)]
[ProducesResponseType(StatusCode.Status500InternalServerError)]
public async Task<ActionResult<IEnumerable<NewsResponseTo>>>
GetAllNews()
    {
        try
        {
            _logger.LogInformation("Getting all news");

            var news = await _newsService.GetAllNews();

            return Ok(news);
        }
    }

```

```

    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error getting all news");
        return StatusCode(500, "An error occurred while retrieving news");
    }
}

[HttpGet("{id:long}")]
[ProducesResponseType(typeof(NewsResponseTo), StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<NewsResponseTo>>
GetNewsById([FromRoute] long id)
{
    try
    {
        _logger.LogInformation("Getting news by id: {Id}", id);

        var getNewsRequest = new NewsRequestTo { Id = id };
        var news = await _newsService.GetNews(getNewsRequest);

        return Ok(news);
    }
    catch (NewNotFoundException ex)
    {
        _logger.LogWarning(ex, "News not found with id: {Id}", id);
        return NotFound($"News with id {id} not found");
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error getting news by id: {Id}", id);
        return StatusCode(500, "An error occurred while retrieving the news");
    }
}

[HttpPut]
[ProducesResponseType(typeof(NewsResponseTo), StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status400BadRequest)]
[ProducesResponseType(StatusCodes.Status404NotFound)]

```

```

[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<NewsResponseTo>> UpdateNews(
    [FromBody] NewsRequestTo updateNewsRequest)
{
    try
    {
        _logger.LogInformation("Updating news with id: {Id}",
updateNewsRequest.Id);

        var updatedNews = await
_newsService.UpdateNews(updateNewsRequest);

        return Ok(updatedNews);
    }
    catch (NewNotFoundException ex)
    {
        _logger.LogWarning(ex, "News not found for update with id: {Id}",
updateNewsRequest.Id);
        return NotFound();
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error updating news with id: {Id}",
updateNewsRequest.Id);
        return StatusCode(500, "An error occurred while updating the news");
    }
}

```

```

[HttpDelete("{id:long}")]
[ProducesResponseType(StatusCodes.Status204NoContent)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<IActionResult> DeleteNews([FromRoute] long id)
{
    try
    {
        _logger.LogInformation("Deleting news with id: {Id}", id);

        var deleteNewsRequest = new NewsRequestTo { Id = id };
        await _newsService.DeleteNews(deleteNewsRequest);
    }
}

```

```

        return NoContent();
    }
    catch (NewNotFoundException ex)
    {
        _logger.LogWarning(ex, "News not found for deletion with id: {Id}", id);
        return NotFound();
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error deleting news with id: {Id}", id);
        return StatusCode(500, "An error occurred while deleting the news");
    }
}
}

```

```

using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

```

```

namespace rest1.api.controllers;

```

```

[ApiController]
[Route("api/v1.0/[controller]")]
public class NotesController : ControllerBase
{
    private readonly INoteService _noteService;
    private readonly ILogger<NotesController> _logger;

    public NotesController(INoteService noteService, ILogger<NotesController>
logger)
    {
        _noteService = noteService;
        _logger = logger;
    }

    [HttpPost]
    [ProducesResponseType(typeof(NoteResponseTo),

```

```

        StatusCodes.Status201Created)]
        [ProducesResponseType(StatusCodes.Status400BadRequest)]
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<ActionResult<NoteResponseTo>> CreateNote(
            [FromBody] NoteRequestTo createNoteRequest)
        {
            try
            {
                _logger.LogInformation("Creating note {request}", createNoteRequest);

                NoteResponseTo createdNote =
                    await _noteService.CreateNote(createNoteRequest);

                return CreatedAtAction(
                    nameof(CreateNote),
                    new { id = createdNote.Id },
                    createdNote
                );
            }
            catch (Exception ex)
            {
                _logger.LogError(ex, "Error creating note");
                return StatusCode(500, "An error occurred while creating the note");
            }
        }

        [HttpGet]
        [ProducesResponseType(typeof(IEnumerable<NoteResponseTo>),
        StatusCodes.Status200OK)]
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<ActionResult<IEnumerable<NoteResponseTo>>>
        GetAllNotes()
        {
            try
            {
                _logger.LogInformation("Getting all notes");

                IEnumerable<NoteResponseTo> notes =
                    await _noteService.GetAllNotes();
            }
        }
    }
}

```

```

        return Ok(notes);
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error getting all notes");
        return StatusCode(500, "An error occurred while retrieving notes");
    }
}

[HttpGet("{id:long}")]
[ProducesResponseType(typeof(NoteResponseTo),
StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<NoteResponseTo>>
GetNoteById([FromRoute] long id)
{
    try
    {
        _logger.LogInformation("Getting note by id: {Id}", id);

        var getNoteRequest = new NoteRequestTo { Id = id };
        NoteResponseTo note =
            await _noteService.GetNote(getNoteRequest);

        return Ok(note);
    }
    catch (NewNotFoundException ex)
    {
        _logger.LogWarning(ex, "Note not found with id: {Id}", id);
        return NotFound($"Note with id {id} not found");
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error getting note by id: {Id}", id);
        return StatusCode(500, "An error occurred while retrieving the note");
    }
}

[HttpPut]

```

```

[ProducesResponseType(typeof(NoteResponseTo),
StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status400BadRequest)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<NoteResponseTo>> UpdateNote(
    [FromBody] NoteRequestTo updateNoteRequest)
{
    try
    {
        _logger.LogInformation("Updating note with id: {Id}",
updateNoteRequest.Id);

        var updatedNote =
            await _noteService.UpdateNote(updateNoteRequest);

        return Ok(updatedNote);
    }
    catch (NewNotFoundException ex)
    {
        _logger.LogWarning(
            ex,
            "Note not found for update with id: {Id}",
            updateNoteRequest.Id);

        return NotFound(
            $"Note with id {updateNoteRequest.Id} not found");
    }
    catch (Exception ex)
    {
        _logger.LogError(
            ex,
            "Error updating note with id: {Id}",
            updateNoteRequest.Id);

        return StatusCode(
            500,
            "An error occurred while updating the note");
    }
}

```

```

[HttpDelete("{id:long}")]
[ProducesResponseType(StatusCodes.Status204NoContent)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<IActionResult> DeleteNote([FromRoute] long id)
{
    try
    {
        _logger.LogInformation("Deleting note with id: {Id}", id);

        var deleteNoteRequest = new NoteRequestTo { Id = id };
        await _noteService.DeleteNote(deleteNoteRequest);

        return NoContent();
    }
    catch (NoteNotFoundException ex)
    {
        _logger.LogWarning(ex, "Note not found for deletion with id: {Id}", id);
        return NotFound();
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error deleting note with id: {Id}", id);
        return StatusCode(500, "An error occurred while deleting the note");
    }
}
}

```

```
using System.ComponentModel.DataAnnotations;
```

```
namespace rest1.application.DTOs.requests;
```

```

public class CreatorRequestTo
{
    public long? Id { get; set; }

    [StringLength(64, MinimumLength = 2)]
    public string? Login { get; set; }
}

```

```

        [StringLength(128, MinimumLength = 8)]
        public string? Password { get; set; }

        [StringLength(64, MinimumLength = 2)]
        public string? Firstname { get; set; }

        [StringLength(64, MinimumLength = 2)]
        public string? Lastname { get; set; }
    }

    using System.ComponentModel.DataAnnotations;

    namespace rest1.application.DTOs.requests;

    public class MarkRequestTo
    {
        public long? Id { get; set; }

        [StringLength(32, MinimumLength = 2)]
        public string? Name { get; set; }

        public long[] NewsIds { get; set; }
    }

    using System.ComponentModel.DataAnnotations;

    namespace rest1.application.DTOs.requests;

    public class NewsRequestTo
    {
        public long? Id { get; set; }

        public long CreatorId { get; set; }

        [StringLength(64, MinimumLength = 2)]
        public string Title { get; set; } = string.Empty;

        [StringLength(2048, MinimumLength = 4)]
        public string Content { get; set; } = string.Empty;
    }

```

```

    public long[] MarkIds { get; set; }
}

using System.ComponentModel.DataAnnotations;

namespace rest1.application.DTOs.requests;

public class NoteRequestTo
{
    public long? Id { get; set; }

    public long NewsId { get; set; }

    [StringLength(2048, MinimumLength = 2)]
    public string Content { get; set; }
}

namespace rest1.application.DTOs.responses;

public class CreatorResponseTo (
    string login,
    string password,
    string firstName,
    string lastName)
{
    public long Id { get; set; }

    public string Login { get; set; } = login;

    public string Password { get; set; } = password;

    public string Firstname { get; set; } = firstName;

    public string Lastname { get; set; } = lastName;
}

namespace rest1.application.DTOs.responses;

public class MarkResponseTo(

```

```

        long id,
        string name)
    {
        public long? Id { get; set; } = id;

        public string? Name { get; set; } = name;
    }

namespace rest1.application.DTOs.responses;

public class NewsResponseTo(
    string title,
    string content,
    DateTime createdAt,
    DateTime modified)
{
    public long Id { get; set; }

    public long CreatorId { get; set; }

    public string Title { get; set; } = title;

    public string Content { get; set; } = content;

    public DateTime CreatedAt { get; set; } = createdAt;

    public DateTime Modified { get; set; } = modified;
}

namespace rest1.application.DTOs.responses;

public class NoteResponseTo(
    long id,
    long newsId,
    string content)
{
    public long Id { get; set; } = id;

    public long NewsId { get; set; } = newsId;

```

```
    public string Content { get; set; } = content;
}
```

```
namespace rest1.application.exceptions;
```

```
public class CreatorNotFoundException : NotFoundException
{
    public CreatorNotFoundException(string message, Exception inner) :
base(message, inner) { }

    public CreatorNotFoundException(string message) : base(message) { }
}
```

```
namespace rest1.application.exceptions;
```

```
public class MarkNotFoundException : NotFoundException
{
    public MarkNotFoundException(string message, Exception inner) :
base(message, inner) { }

    public MarkNotFoundException(string message) : base(message) { }
}
```

```
namespace rest1.application.exceptions;
```

```
public class NewNotFoundException : NotFoundException
{
    public NewNotFoundException(string message, Exception inner) :
base(message, inner) { }

    public NewNotFoundException(string message) : base(message) { }
}
```

```
namespace rest1.application.exceptions;
```

```
public class NotFoundException : Exception
{
    public NotFoundException(string message, Exception inner) : base(message,
inner) { }
```

```

    public NotFoundException(string message) : base(message) { }
}

namespace rest1.application.exceptions;

public class NoteNotFoundException : NotFoundException
{
    public NoteNotFoundException(string message, Exception inner) :
base(message, inner) { }

    public NoteNotFoundException(string message) : base(message) { }
}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface ICreatorRepository : IRepository<Creator>
{

}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface IMarkRepository : IRepository<Mark>
{

}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface INewsRepository : IRepository<News>
{

}

```

```

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface INoteRepository : IRepository<Note>
{
}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface IRepository<TEntity> where TEntity : Entity
{
    //get records
    Task<TEntity?> GetByIdAsync(long id, CancellationToken cancellationToken
= default);
    Task<IEnumerable<TEntity>> GetAllAsync(CancellationToken ct = default);

    //create record
    Task<TEntity> AddAsync(TEntity entity, CancellationToken ct = default);

    //update record
    Task<TEntity?> UpdateAsync(TEntity entity, CancellationToken ct = default);

    //delete record
    Task<TEntity?> DeleteAsync(TEntity entity, CancellationToken ct = default);

    //find records
    Task<TEntity?> FindAsync(ISpecification<TEntity> spec, CancellationToken
ct = default);
    Task<IEnumerable<TEntity>> FindAllAsync(ISpecification<TEntity> spec,
CancellationToken ct = default);

    //count records
    Task<int> CountAsync(ISpecification<TEntity> spec, CancellationToken ct =
default);

    //is any record

```

```
    Task<bool> AnyAsync(ISpecification<TEntity> spec, CancellationToken ct =
default);
}
```

```
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
```

```
namespace rest1.application.interfaces.services;
```

```
public interface ICreatorService
{
```

```
    Task<CreatorResponseTo> CreateCreator(CreatorRequestTo
createCreatorRequestTo);
```

```
    Task<IEnumerable<CreatorResponseTo>> GetAllCreators();
```

```
    Task<CreatorResponseTo> GetCreator(CreatorRequestTo
getCreatorRequestTo);
```

```
    Task<CreatorResponseTo> UpdateCreator(CreatorRequestTo
updateCreatorRequestTo);
```

```
    Task DeleteCreator(CreatorRequestTo deleteCreatorRequestTo);
}
```

```
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
```

```
namespace rest1.application.interfaces.services;
```

```
public interface IMarkService
{
```

```
    Task<MarkResponseTo> CreateMark(MarkRequestTo createMarkRequestTo);
```

```
    Task<IEnumerable<MarkResponseTo>> GetAllMarks();
```

```
    Task<MarkResponseTo> GetMark(MarkRequestTo getMarkRequestTo);
```

```
    Task<MarkResponseTo> UpdateMark(MarkRequestTo updateMarkRequestTo);
```

```

    Task DeleteMark(MarkRequestTo deleteMarkRequestTo);
}

using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;

namespace rest1.application.interfaces.services;

public interface INewsService
{
    Task<NewsResponseTo> CreateNews(NewsRequestTo createNewsRequestTo);

    Task<IEnumerable<NewsResponseTo>> GetAllNews();

    Task<NewsResponseTo> GetNews(NewsRequestTo getNewsRequestTo);

    Task<NewsResponseTo> UpdateNews(NewsRequestTo
updateNewsRequestTo);

    Task DeleteNews(NewsRequestTo deleteNewsRequestTo);
}

using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;

namespace rest1.application.interfaces.services;

public interface INoteService
{
    Task<NoteResponseTo> CreateNote(NoteRequestTo createNoteRequestTo);

    Task<IEnumerable<NoteResponseTo>> GetAllNotes();

    Task<NoteResponseTo> GetNote(NoteRequestTo getNoteRequestTo);

    Task<NoteResponseTo> UpdateNote(NoteRequestTo updateNoteRequestTo);

    Task DeleteNote(NoteRequestTo deleteNoteRequestTo);
}

```

```

using System.Linq.Expressions;

namespace rest1.application.interfaces;

public interface ISpecification<T>
{
    //criteria expression
    Expression<Func<T, bool>> Criteria { get; }

    //join expressions
    List<Expression<Func<T, object>>> Includes { get; }

    //order by expressions
    Expression<Func<T, object>>? OrderBy { get; }
    Expression<Func<T, object>>? OrderByDescending { get; }

    //taken records count
    int? Take { get; }

    //skipped records count
    int? Skip { get; }

    //pagination flag
    bool IsPagingEnabled { get; }
}

using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;
using AutoMapper;

namespace rest1.application.mappers;

public class CreatorProfile : Profile
{
    public CreatorProfile()
    {
        CreateMap<CreatorRequestTo, Creator>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.Login, opt => opt.MapFrom(src => src.Login))
    }
}

```

```

        .ForMember(dest => dest.Firstname, opt => opt.MapFrom(src =>
src.Firstname))
        .ForMember(dest => dest.Lastname, opt => opt.MapFrom(src =>
src.Lastname))
        .ForMember(dest => dest.Password, opt => opt.MapFrom(src =>
src.Password));

    CreateMap<Creator, CreatorResponseTo>()
        .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
        .ForMember(dest => dest.Login, opt => opt.MapFrom(src => src.Login))
        .ForMember(dest => dest.Firstname, opt => opt.MapFrom(src =>
src.Firstname))
        .ForMember(dest => dest.Lastname, opt => opt.MapFrom(src =>
src.Lastname))
        .ForMember(dest => dest.Password, opt => opt.MapFrom(src =>
src.Password));
    }
}

```

```

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;

```

```

namespace rest1.application.mappers;

```

```

public class MarkProfile : Profile
{
    public MarkProfile()
    {
        CreateMap<MarkRequestTo, Mark>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.Name, opt => opt.MapFrom(src => src.Name));

        CreateMap<Mark, MarkResponseTo>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
            .ForMember(dest => dest.Name, opt => opt.MapFrom(src => src.Name));
    }
}

```

```

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;

namespace rest1.application.mappers;

public class NewsProfile : Profile
{
    public NewsProfile()
    {
        CreateMap<NewsRequestTo, News>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.CreatedAt, opt => opt.MapFrom(src =>
DateTime.UtcNow))
            .ForMember(dest => dest.Modified, opt => opt.MapFrom(src =>
DateTime.UtcNow))
            .ForMember(dest => dest.CreatorId, opt => opt.MapFrom(src =>
src.CreatorId))
            .ForMember(dest => dest.Content, opt => opt.MapFrom(src =>
src.Content))
            .ForMember(dest => dest.Title, opt => opt.MapFrom(src => src.Title));

        CreateMap<News, NewsResponseTo>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
            .ForMember(dest => dest.CreatorId, opt => opt.MapFrom(src =>
src.CreatorId))
            .ForMember(dest => dest.Title, opt => opt.MapFrom(src => src.Title))
            .ForMember(dest => dest.Content, opt => opt.MapFrom(src =>
src.Content))
            .ForMember(dest => dest.CreatedAt, opt => opt.MapFrom(src =>
src.CreatedAt))
            .ForMember(dest => dest.Modified, opt => opt.MapFrom(src =>
src.Modified));
    }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;

```

```

using rest1.core.entities;

namespace rest1.application.mappers;

public class NoteProfile : Profile
{
    public NoteProfile()
    {
        CreateMap<NoteRequestTo, Note>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.NewsId, opt => opt.MapFrom(src =>
src.NewsId));

        CreateMap<Note, NoteResponseTo>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
            .ForMember(dest => dest.NewsId, opt => opt.MapFrom(src =>
src.NewsId));
    }
}

```

```

using AutoMapper;
using Microsoft.AspNetCore.Components.Forms;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;

```

```

namespace rest1.application.services;

```

```

public class CreatorService : ICreatorService
{
    private readonly IMapper _mapper;
    private readonly ICreatorRepository _creatorRepository;

    public CreatorService(IMapper mapper, ICreatorRepository repository)
    {
        _mapper = mapper;
        _creatorRepository = repository;
    }
}

```

```

    }

    public async Task<CreatorResponseTo> CreateCreator(CreatorRequestTo
createCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(createCreatorRequestTo);

        Creator createdCreator = await
        _creatorRepository.AddAsync(creatorFromDto);

        CreatorResponseTo dtoFromCreatedEditor =
        _mapper.Map<CreatorResponseTo>(createdCreator);

        return dtoFromCreatedEditor;
    }

    public async Task DeleteCreator(CreatorRequestTo deleteCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(deleteCreatorRequestTo);

        _ = await _creatorRepository.DeleteAsync(creatorFromDto)
        ?? throw new CreatorNotFoundException(
            $"Delete creator {creatorFromDto} was not found");
    }

    public async Task<IEnumerable<CreatorResponseTo>> GetAllCreators()
    {
        IEnumerable<Creator> allCreators =
            await _creatorRepository.GetAllAsync();

        var response = new List<CreatorResponseTo>();

        foreach (Creator creator in allCreators)
        {
            response.Add(_mapper.Map<CreatorResponseTo>(creator));
        }

        return response;
    }

```

```

    public async Task<CreatorResponseTo> GetCreator(CreatorRequestTo
getCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(getCreatorRequestTo);

        Creator demandedCreator =
            await _creatorRepository.GetByIdAsync(creatorFromDto.Id)
            ?? throw new CreatorNotFoundException(
                $"Demanded creator {creatorFromDto} was not found");

        return _mapper.Map<CreatorResponseTo>(demandedCreator);
    }

    public async Task<CreatorResponseTo> UpdateCreator(CreatorRequestTo
updateCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(updateCreatorRequestTo);

        Creator updatedCreator =
            await _creatorRepository.UpdateAsync(creatorFromDto)
            ?? throw new CreatorNotFoundException(
                $"Update creator {creatorFromDto} was not found");

        return _mapper.Map<CreatorResponseTo>(updatedCreator);
    }
}

```

```

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;

```

```

namespace rest1.application.services;

```

```

public class MarkService : IMarkService
{
    private readonly IMapper _mapper;

```

```

private readonly IMarkRepository _markRepository;

public MarkService(IMapper mapper, IMarkRepository repository)
{
    _mapper = mapper;
    _markRepository = repository;
}

public async Task<MarkResponseTo> CreateMark(MarkRequestTo
createMarkRequestTo)
{
    Mark markFromDto = _mapper.Map<Mark>(createMarkRequestTo);

    Mark createdMark = await _markRepository.AddAsync(markFromDto);

    MarkResponseTo dtoFromCreatedMark =
        _mapper.Map<MarkResponseTo>(createdMark);

    return dtoFromCreatedMark;
}

public async Task DeleteMark(MarkRequestTo deleteMarkRequestTo)
{
    Mark markFromDto = _mapper.Map<Mark>(deleteMarkRequestTo);

    _ = await _markRepository.DeleteAsync(markFromDto)
        ?? throw new MarkNotFoundException(
            $"Delete mark {markFromDto} was not found");
}

public async Task<IEnumerable<MarkResponseTo>> GetAllMarks()
{
    IEnumerable<Mark> allMarks =
        await _markRepository.GetAllAsync();

    var allMarksResponseTos = new List<MarkResponseTo>();

    foreach (Mark mark in allMarks)
    {
        MarkResponseTo markTo =

```

```

        _mapper.Map<MarkResponseTo>(mark);

        allMarksResponseTos.Add(markTo);
    }

    return allMarksResponseTos;
}

public async Task<MarkResponseTo> GetMark(MarkRequestTo
getMarksRequestTo)
{
    Mark markFromDto = _mapper.Map<Mark>(getMarksRequestTo);

    Mark demandedMark =
        await _markRepository.GetByIdAsync(markFromDto.Id)
        ?? throw new MarkNotFoundException(
            $"Demanded mark {markFromDto} was not found");

    MarkResponseTo demandedMarkResponseTo =
        _mapper.Map<MarkResponseTo>(demandedMark);

    return demandedMarkResponseTo;
}

public async Task<MarkResponseTo> UpdateMark(MarkRequestTo
updateMarkRequestTo)
{
    Mark markFromDto = _mapper.Map<Mark>(updateMarkRequestTo);

    Mark updateMark =
        await _markRepository.UpdateAsync(markFromDto)
        ?? throw new MarkNotFoundException(
            $"Update mark {markFromDto} was not found");

    MarkResponseTo updateMarkResponseTo =
        _mapper.Map<MarkResponseTo>(updateMark);

    return updateMarkResponseTo;
}
}

```

```
using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;
```

```
namespace rest1.application.services;
```

```
public class NewsService : INewsService
{
    private readonly IMapper _mapper;

    private readonly INewsRepository _newsRepository;

    public NewsService(IMapper mapper, INewsRepository repository)
    {
        _mapper = mapper;
        _newsRepository = repository;
    }

    public async Task<NewsResponseTo> CreateNews(NewsRequestTo
createNewsRequestTo)
    {
        News newsFromDto = _mapper.Map<News>(createNewsRequestTo);

        News createdNews = await _newsRepository.AddAsync(newsFromDto);

        NewsResponseTo dtoFromCreatedNews =
_mapper.Map<NewsResponseTo>(createdNews);

        return dtoFromCreatedNews;
    }

    public async Task DeleteNews(NewsRequestTo deleteNewsRequestTo)
    {
        News newsFromDto = _mapper.Map<News>(deleteNewsRequestTo);
        _ = await _newsRepository.DeleteAsync(newsFromDto) ?? throw new
```

```

NewNotFoundException($"Deleted news {newsFromDto} was not found");
    }

    public async Task<IEnumerable<NewsResponseTo>> GetAllNews()
    {
        IEnumerable<News> allNews = await _newsRepository.GetAllAsync();

        var allNewsResponseTos = new List<NewsResponseTo>();
        foreach (News news in allNews)
        {
            NewsResponseTo newsTo = _mapper.Map<NewsResponseTo>(news);
            allNewsResponseTos.Add(newsTo);
        }

        return allNewsResponseTos;
    }

    public async Task<NewsResponseTo> GetNews(NewsRequestTo
getNewsRequestTo)
    {
        News newsFromDto = _mapper.Map<News>(getNewsRequestTo);

        News demandedNews = await
        _newsRepository.GetByIdAsync(newsFromDto.Id) ?? throw new
        NewNotFoundException($"Demanded news {newsFromDto} was not found");

        NewsResponseTo demandedNewsResponseTo =
        _mapper.Map<NewsResponseTo>(demandedNews);

        return demandedNewsResponseTo;
    }

    public async Task<NewsResponseTo> UpdateNews(NewsRequestTo
updateNewsRequestTo)
    {
        News newsFromDto = _mapper.Map<News>(updateNewsRequestTo);

        News? updateNews = await _newsRepository.UpdateAsync(newsFromDto)
        ?? throw new NewNotFoundException($"Update news {newsFromDto} was not
        found");
    }

```

```
        NewsResponseTo updateNewsResponseTo =  
_mapper.Map<NewsResponseTo>(updateNews);
```

```
        return updateNewsResponseTo;  
    }  
}
```

```
using AutoMapper;  
using rest1.application.DTOs.requests;  
using rest1.application.DTOs.responses;  
using rest1.application.exceptions;  
using rest1.application.interfaces;  
using rest1.application.interfaces.services;  
using rest1.core.entities;
```

```
namespace rest1.application.services;
```

```
public class NoteService : INoteService
```

```
{  
    private readonly IMapper _mapper;  
    private readonly INoteRepository _noteRepository;
```

```
    public NoteService(IMapper mapper, INoteRepository repository)  
    {  
        _mapper = mapper;  
        _noteRepository = repository;  
    }
```

```
    public async Task<NoteResponseTo> CreateNote(NoteRequestTo  
createNoteRequestTo)  
    {  
        Note noteFromDto = _mapper.Map<Note>(createNoteRequestTo);  
  
        Note createdNote = await _noteRepository.AddAsync(noteFromDto);  
  
        NoteResponseTo dtoFromCreatedNote =  
            _mapper.Map<NoteResponseTo>(createdNote);  
  
        return dtoFromCreatedNote;
```

```

    }

    public async Task DeleteNote(NoteRequestTo deleteNoteRequestTo)
    {
        Note noteFromDto = _mapper.Map<Note>(deleteNoteRequestTo);

        _ = await _noteRepository.DeleteAsync(noteFromDto)
            ?? throw new NoteNotFoundException(
                $"Delete note {noteFromDto} was not found");
    }

    public async Task<IEnumerable<NoteResponseTo>> GetAllNotes()
    {
        IEnumerable<Note> allNotes =
            await _noteRepository.GetAllAsync();

        var response = new List<NoteResponseTo>();

        foreach (Note note in allNotes)
        {
            response.Add(_mapper.Map<NoteResponseTo>(note));
        }

        return response;
    }

    public async Task<NoteResponseTo> GetNote(NoteRequestTo
getNoteRequestTo)
    {
        Note noteFromDto = _mapper.Map<Note>(getNoteRequestTo);

        Note demandedNote =
            await _noteRepository.GetByIdAsync(noteFromDto.Id)
            ?? throw new NoteNotFoundException(
                $"Demanded note {noteFromDto} was not found");

        return _mapper.Map<NoteResponseTo>(demandedNote);
    }

    public async Task<NoteResponseTo> UpdateNote(NoteRequestTo

```

```

updateNoteRequestTo)
{
    Note noteFromDto = _mapper.Map<Note>(updateNoteRequestTo);

    Note updatedNote =
        await _noteRepository.UpdateAsync(noteFromDto)
        ?? throw new NoteNotFoundException(
            $"Update note {noteFromDto} was not found");

    return _mapper.Map<NoteResponseTo>(updatedNote);
}

```

```

using System.Linq.Expressions;
using rest1.application.interfaces;

```

```

namespace rest1.application.specifications;

```

```

public abstract class BaseSpecification<T> : ISpecification<T>
{
    public Expression<Func<T, bool>> Criteria { get; private set; }

    public List<Expression<Func<T, object>>> Includes { get; } = [];

    public Expression<Func<T, object>>? OrderBy { get; private set; }

    public Expression<Func<T, object>>? OrderByDescending { get; private set; }

    public int? Take { get; private set; }

    public int? Skip { get; private set; }

    public bool IsPagingEnabled { get; private set; }

    protected BaseSpecification(Expression<Func<T, bool>> criteria)
    {
        Criteria = criteria;
    }

    protected void AddInclude(Expression<Func<T, object>> includeExpression)

```

```

    {
        Includes.Add(includeExpression);
    }

    protected void ApplyPaging(int skip, int take)
    {
        Skip = skip;
        Take = take;
        IsPagingEnabled = true;
    }

    protected void ApplyOrderBy(Expression<Func<T, object>>
orderByExpression)
    {
        OrderBy = orderByExpression;
    }

    protected void ApplyOrderByDescending(Expression<Func<T, object>>
orderByDescendingExpression)
    {
        OrderByDescending = orderByDescendingExpression;
    }
}

namespace rest1.core.entities;

public class Creator(
    string login,
    string password,
    string firstname,
    string lastname) : Entity
{
    public string Login { get; set; } = login;

    public string Password { get; set; } = password;

    public string Firstname { get; set; } = firstname;

    public string Lastname { get; set; } = lastname;
}

```

```
namespace rest1.core.entities;
```

```
public abstract class Entity  
{  
    public long Id { get; set; }  
}
```

```
namespace rest1.core.entities;
```

```
public class Mark(string name) : Entity  
{  
    public string Name { get; set; } = name;  
    public long[] NewsIds { get; set; } = new long[0];  
}
```

```
namespace rest1.core.entities;
```

```
public class News : Entity  
{  
    public long CreatorId { get; set; }  
  
    public string Title { get; set; } = string.Empty;  
  
    public string Content { get; set; } = string.Empty;  
  
    public DateTime CreatedAt { get; set; }  
  
    public DateTime Modified { get; set; }  
  
    // lookup object  
    public Creator? Creator { get; set; }  
    public long[]? MarkIds { get; set; }  
}
```

```
namespace rest1.core.entities;
```

```
public class Note : Entity  
{  
    public long NewsId { get; set; }  
}
```

```

    public string Content { get; set; } = string.Empty;

    // lookup object
    public News? News { get; set; }

}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class CreatorRepository : Repository<Creator>, ICreatorRepository
{

}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class MarkRepository: Repository<Mark>, IMarkRepository
{

}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class NewsRepository: Repository<News>, INewsRepository
{

}

using rest1.application.interfaces;

```

```

using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class NoteRepository: Repository<Note>, INoteRepository
{

}

using System.Collections.Concurrent;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class Repository<TEntity> : IRepository<TEntity> where TEntity : Entity
{
    protected readonly ConcurrentDictionary<long, TEntity> _entities = new();

    private long _nextId = 1;

    private readonly Lock _lock = new();

    public virtual Task<TEntity?> FindAsync(ISpecification<TEntity>
specification, CancellationToken cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            IQueryable<TEntity> query = ApplySpecification(specification);
            return Task.FromResult(query.FirstOrDefault());
        }
    }

    public virtual Task<IEnumerable<TEntity>>
FindAllAsync(ISpecification<TEntity> specification, CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();
    }
}

```

```

lock (_lock)
{
    IQueryable<TEntity> query = ApplySpecification(specification);
    return Task.FromResult(query.ToList().AsEnumerable());
}
}

```

```

public virtual Task<int> CountAsync(ISpecification<TEntity> specification,
CancellationTokens cancellationTokens = default)
{
    cancellationTokens.ThrowIfCancellationRequested();

    lock (_lock)
    {
        IQueryable<TEntity> query = ApplySpecification(specification);
        return Task.FromResult(query.Count());
    }
}

```

```

public virtual Task<bool> AnyAsync(ISpecification<TEntity> specification,
CancellationTokens cancellationTokens = default)
{
    cancellationTokens.ThrowIfCancellationRequested();

    lock (_lock)
    {
        IQueryable<TEntity> query = ApplySpecification(specification);
        return Task.FromResult(query.Any());
    }
}

```

```

// CRUD
public virtual Task<TEntity> AddAsync(TEntity entity, CancellationTokens
cancellationTokens = default)
{
    ArgumentNullException.ThrowIfNull(entity);
    cancellationTokens.ThrowIfCancellationRequested();
}

```

```

lock (_lock)
{
    var id = entity.Id;
    if (id == 0)
    {
        id = _nextId;
    }

    if (_entities.TryAdd(_nextId, entity))
    {
        entity.Id = id;
        _nextId++;
        return Task.FromResult(entity);
    }

    throw new InvalidOperationException($"Entity with id {id} already
exists");
}

}

public virtual Task<TEntity?> UpdateAsync(TEntity entity, CancellationToken
cancellation_token = default)
{
    ArgumentNullException.ThrowIfNull(entity);
    cancellation_token.ThrowIfCancellationRequested();

    lock (_lock)
    {
        var id = entity.Id;

        if (_entities.TryGetValue(id, out TEntity? updateEntity))
        {
            _entities[id] = entity;
            updateEntity = _entities[id];
        }

        return Task.FromResult(updateEntity);
    }
}

```

```

    public virtual Task<TEntity?> DeleteAsync(TEntity entity, CancellationToken
cancellationToken = default)
    {
        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            var id = entity.Id;
            _entities.TryRemove(id, out TEntity? deletedEntity);

            return Task.FromResult(deletedEntity);
        }
    }

```

```

    public virtual Task<bool> DeleteByIdAsync(long id, CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            return Task.FromResult(_entities.TryRemove(id, out _));
        }
    }

```

```

    public virtual Task<TEntity?> GetByIdAsync(long id, CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            return Task.FromResult(_entities.GetValueOrDefault(id));
        }
    }

```

```

    public virtual Task<IEnumerable<TEntity>> GetAllAsync(CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();
    }

```

```

        lock (_lock)
        {
            return Task.FromResult(_entities.Values.AsEnumerable());
        }
    }
    protected virtual IQueryable<TEntity>
    ApplySpecification(ISpecification<TEntity> spec)
    {
        IQueryable<TEntity> query = _entities.Values.AsQueryable();

        if (spec.Criteria != null)
        {
            query = query.Where(spec.Criteria);
        }
        if (spec.OrderBy != null)
        {
            query = query.OrderBy(spec.OrderBy);
        }
        else if (spec.OrderByDescending != null)
        {
            query = query.OrderByDescending(spec.OrderByDescending);
        }

        if (spec.IsPagingEnabled)
        {
            query = query.Skip(spec.Skip ?? 0)
                .Take(spec.Take ?? 10);
        }
        return query;
    }
    public void Clear()
    {
        lock (_lock)
        {
            _entities.Clear();
        }
    }
}

```